Design and Implementation of College Student Management System Based on Computer BS Architecture

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Abstract

To enhance understanding of the design and implementation of a student management system tailored for the Higher People's Court, a study on the development of such a system using the computer-based B/S (Browser/Server) architecture is proposed. This research focuses on leveraging database and related technologies to construct a comprehensive college student information management system. By adopting the B/S structure, the system significantly advances the informatization of student data management processes.The paper presents an information management framework specifically designed for managing college student records, highlighting its primary objectives, key design elements, and operational workflows. Additionally, it examines the user interaction pathways to optimize the system's functionality and overall performance, thereby improving its efficiency and practicality in college information management scenarios. Finally, the practice proves that the system has good security, reliability and expansibility, which provides a strong guarantee for the management of school students. The system has achieved the expected results from the construction to the operation of the management system, which provides a good platform for improving the efficiency of school student management and reducing management costs.

CCS Concepts

• **Networks** → Network services; Network management.

Keywords

computer BS architecture, Information management, system design

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1 Introduction

In recent years, the steady growth in college enrollment has led to a significant increase in student populations, making student management a critical aspect of daily operations in higher education institutions. As enrollment numbers rise, students are drawn from increasingly diverse backgrounds, and their academic abilities vary widely, prompting universities to establish more robust requirements for effective management. Traditional methods of managing student information have become inadequate to address these evolving needs. In response, a student information management system utilizing the computer-based B/S (Browser/Server) architecture has been developed to streamline and modernize student management processes, which provides a convenient, fast and interactive student information management platform and can realize all-round management of students. The system has the function of inquiring about students' corresponding information, and can be used for managers and operators to manage students' information at different levels of authority, which is convenient for operators to add, inquire and modify information at any time. The development of this system is based on the convenience of students and teachers and reducing their labor intensity, which saves time for teachers and managers, and also saves management costs for the school. The system is designed to be highly reliable, secure, and cost-effective, significantly enhancing the efficiency of managing student information. By enabling detailed and systematic analysis of student data, it offers a fast and user-friendly platform for the dissemination of information by the student management department. This not only boosts the operational efficiency and service standards of the institution's student management but also provides valuable data-driven insights to support informed decision-making across various school functions [1-2].

2 System requirements analysis

The student management system based on computer BS architecture is mainly based on student information management, supplemented by students' basic information, achievement information and file information, and comprehensively manages students. Its functions include:

(1) Data maintenance function: maintain students' personal data and class data, and realize user management, user function management and other functions.

(2) User management function: the basic information of users can be maintained and modified according to the authority of the administrator.

(3) Data query function: query the data provided by users, mainly including personal information query and class information query.

Table 1: Design of College Student Information Management System

Student management structure	Student learning management system	Basic information word structure	query
Teacher management structure	database	Teacher management substructure	modify

(4) Statistical analysis function: it provides the function of statistical analysis of data, including statistics of students' basic personal information, basic class situation and comprehensive class situation.

(5) System management function: providing basic information and authority management for system administrators.

2.1 Database design

The design of the database is a fundamental aspect of system development. Creating a database that is well-structured, efficient, secure, and dependable requires careful consideration of several key factors:

1. Design of data model. The data model is the skeleton of the database, which organizes the data according to a specific logical relationship, thus ensuring the interdependence between the data.

2. Table structure design is a concrete description of database tables, which reflects the relationship between data tables, that is, what type of data is used to organize it.

3. Index design. Index refers to the relationship between one table and another table when establishing some special relationships in the database, which can change the field types and field attribute values of other tables.

4. The design of stored procedure, which refers to the classified storage of data, so that applications can easily obtain information from it.

5. Stored procedures have nothing to do with the data source, and they can be passed to the data source through some mechanism as long as necessary.

2.2 Overall system design

The computer-based B/S architecture student management system is composed of four main components: the data collection layer, the data handling layer, the business logic layer, and the user interface layer. The data acquisition layer mainly includes the login and exit of system administrators, the maintenance of students' basic information, and the query of statistical reports of students' management business. The data processing layer mainly includes college data management center, educational administration information center and examination management center, which is mainly responsible for the basic operation and daily business of the system. The business logic layer mainly includes the module of maintaining students' basic information, the module of maintaining students' reward and punishment information and the module of maintaining graduates' information, which is mainly responsible for the basic business functions of the system. The performance layer mainly includes teachers, academic staff and individual students. The presentation layer is the foreground part of the system.

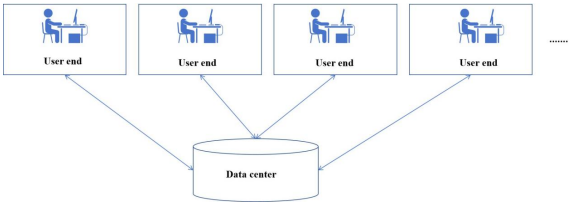

Figure 1: System Structure Diagram

Users access it through a Web browser and interact with the server directly [3-4].

The system adopts three-tier B/S architecture, which realizes the role transformation from users to system administrators and ordinary users. Through this system, the college can effectively manage students' basic information, which is convenient for teachers and student managers to query and modify students' basic information, and improves work efficiency and quality. As shown in Table 1

2.3 B/S System Architecture

The entire system adopts browser/server mode, and the network support platform selects DELLR730 Server (E5-2640V4 * 2/32 G/600G10 KSAS2.5 * 5/H730P), Raid5/750 W*2 Mbit/s), the database adopts SQL Server 2012; The user end is a personal computer, operating system is Microsoft Windows 7, and browser is Microsoft 8.0 or above. The entire system is using Developed by. net. Figure 1 shows the system architecture diagram.

2.4 System Hardware Design Based on B/S Framework

Based on the architecture of B/S, there are 3 kinds of human communication interfaces: the document interface, the video operation interface, and the scene editor interface. Among them, the GUI is used to load and store the DVD or DVD, the VCI is to regulate and deal with the VCI in the course of the PC. The hardware component of CAD based on B/S frame is composed of visual interactive sub-system, graphical processing module, controller module and executing module.

(1) The Visual Interactive Sub-System is in charge of the implementation of all kinds of visual design on the basis of the system node. The VPF is developed as a relational tool, and is shown in a graphic data-stream format. Abstract node is a general type of fundamental control, and it defines the general properties of computer aided system. Abstract nodes are represented by a graphical interface. At the time of creation of every control, the system reads out the location information of the controller according to the inner message, the number of the interface, and the control note,

and outputs the location of the connecting terminal. The VPPROCESS model is used in the control of the handling class. Processor control is associated with a variety of I/O data names and types. The concrete content of this control consists of the primary region, the entry and exit of the data, and the scope of the control can be adjusted by the length of the heading and the title.

(2) Graphical handling module offers the interface and visualization level for the CAD system. A graphical processor consists of a scene, a view, and a graph. There are many graphic elements in a scene, and many of them can be combined into one. In CAD software, the CAD system can be used to implement a variety of functions, which can be accomplished by agent windows, which can offer the best time to interact visually. Graph processing module also has the ability of drawing scene, so that more than one view can be connected with one scene, so that the user can easily browse and utilize a wide variety of scenarios. Furthermore, this module can be used to interact with and show the CAD system, which is a good example of CAD.

(3) The control module combines a strong and user-oriented control toolkit for comprehensive management of all kinds of control. The concrete control approach is to adopt the pattern of the mapping object. The controller will transmit commands to the assistant system when it is required to create a job or an object, which will then generate an internal object and return it to the user who has made the request. Object Value Pattern does not have a fixed pattern, such as a character, number, or picture type. Various kinds of object values are stored in the mapping system. The controller can give some feedback when the user chooses one kind of node, and the class of the system builder is decided by the total quantity of the whole variable.

(4) The structure of executing module is identical to that of B/S. Architecture, which consists of interface and logical level. The function of the interface layer is to realize the control vision. Logical level defines the logic relations among the nodes, and offers a kind of visualization logic programming approach.

3 Application examples

This paper takes a university located in the northwest as an example. The student management system of this university is designed based on B/S architecture and developed by Java language. The system is applied to the management of students in this school, including: freshmen registration, student status information inquiry, graduate employment tracking management and other functions [5-6].

3.1 System functional structure design

During the data collection process, we conducted performance tests on the student management systems of multiple universities, specifically targeting functional modules such as database access, user password modification, addition of student status information, modification of academic performance information, and deletion of disciplinary violations. By simulating different numbers of concurrent users (20, 40, 60, 80, and 100 respectively) and running the system under various network conditions, corresponding response time data samples were obtained. In terms of sample selection, two sets of university student information management systems with similar scale and functionality were selected, namely based on NET architecture designed student information classification management system and comparison system. The sample mainly covers typical university management scenarios and ensures that the hardware configuration and network conditions in the testing environment are as consistent as possible to ensure the comparability of test results.

In terms of data analysis methods, a combination of vertical and horizontal comparisons was adopted. Vertical comparison is the analysis of the relationship between concurrency and response time by observing the impact of different concurrent user numbers on system response time. Horizontal comparison is a comparison of the performance of two systems under the same task, evaluating the performance differences in operations such as information addition and modification. Through these data analysis methods, we can comprehensively evaluate the operational efficiency of the system and verify whether its design performance is superior to the compared system.

Using the JSP technique and the Java developing language, the MySQL database is built. The system's function architecture is illustrated in Table 2.

The system structure is divided into the following parts according to user roles: system administrator, information administrator, students and off-campus users. Among them, the system administrator is responsible for the database management and user information management of the system; The information administrator is responsible for inputting and importing student file information, and managing and maintaining student file information; Student users can query their own file information; Off-campus users are allowed to query the specified student file information [7-8].

3.2 System Interface Design

Using B/S framework, the Student Archives Information Management System is developed by using developing model in accordance with user's function. In terms of interface design, the unified style interface design is adopted, and all interfaces are optimized based on the principles of friendliness, simplicity and convenience, so as to facilitate users' use. Among all user types, the interface adopts a unified style, but the operation type and operation authority are different. The administrator user has the highest authority and all functions can be operated by the administrator user. Student management personnel have most functions, including input, editing, import and deletion, as well as managing the user rights of the class teacher and assigning the query function rights of off-campus personnel and so on.

The login interface adopts unified authentication, and according to the user type selected by the user, after the authenticated user passes, the user enters the operation function modules with different permissions. The user authentication interface is shown in Figure 2.

On the login interface, the user only needs to input the user name and password, without selecting the user type, and the user type is read and judged from the database by the program code, which saves the steps of user operation and improves the friendliness of the user interface. In addition, in the user login interface, the design of verification code is not adopted, but the design of system security is also paid attention to. It not only ensures the security of

Table 2: Functions of Student Archives Information Management System

system administrator	Information administrator	Student users	Off-campus users
database administration	Enter file information	register	register
user management	Import file information	log in	log in
system setup	Manage file information	Inquire personal files	Query the specified file

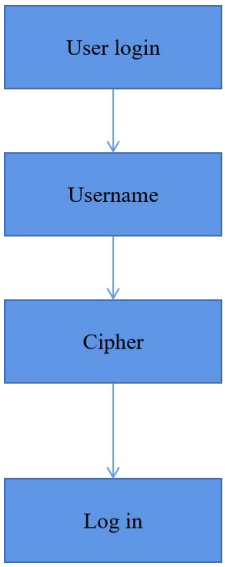

Figure 2: User login

the system, but also saves the steps for users to input verification codes.

Users can choose to remember the password or not when logging in. Remember the password by writing it into the client Cookies, and then reading it from the client Cookies the next time you log in. Forgotten password option is to let the user reset the password by entering the user's authentication information. However, users are advised to remember their passwords unless they choose to use their personal computers, otherwise they will be easily illegally used by others, which will have an impact on system security. The information administrator is the core user who is responsible for the management of student file information, and can input or import student file information [9].

The purpose of this interface design is to allow administrators, users or student managers to input student file information, including students' basic information and daily performance information. Student management personnel can input student information or import student information. This system provides the function of importing students' file information in Excel format, that is, after the students' head teachers have counted the students' file information, they will summarize it in Excel format, and the students' administrators will import it in the background, including basic data information and photos. The function of importing students' information is to summarize Excel, upload the Excel file to the server address at the client of the student management personnel, and read the Excel file by the program code and write it into the MySQL database system. Therefore, when deploying this system, it is necessary to provide the system with read and write permissions on the server side.

In the management of student file information, in addition to the basic text information, it often includes photo information, paper file information scanning files, and may also have some audio and video information. Therefore, the system manages and maintains this part of information in the form of attachments, and the photo information is uploaded to the student's photo location. The paper file is used as an attachment, which users can download when they query, and the audio and video file information is also used as an attachment to provide query and download functions.

Student administrators can also manage and maintain student file information. Including modifying and deleting information. For example, when the student file information changes, the class teacher can apply, and after the relevant leaders pass the examination and approval, the student management personnel will modify the file information. After students leave school, they can archive and delete the file information.

At the same time, for the student management personnel, it also has the function of managing the information of the class teacher and the personnel of the off-campus employment unit. The application is made by the student employment unit, and the student administrator can add the corresponding user information, so that the off-campus personnel can log in to the system and query the corresponding student file information. Therefore, when designing business logic, there is a relationship between the authority of off-campus personnel and the information of student files that can be queried. Class teacher users can input the student file information of their own classes, and also have the function of importing the student file information of their own classes. The authority of the head teacher user includes inputting, importing, modifying and deleting the file information of the students in the class [10].

4 Results and Discussion

This article categorizes the tasks of management functions, including accessing databases, changing user passwords, adding student enrollment information, updating student grade information, and deleting student disciplinary records. Through statistical analysis of these five management tasks and their execution results, it was found that the designed approach is based on The university student information classification management system based on the. NET framework can achieve the expected output results. The average response time for management tasks is 0.55 seconds when no other users are online. Subsequently, 20, 40, 60, 80, and 100 concurrent users were set up, and the response time of the two student information management systems was statistically analyzed under

Table 3: Performance Test Results of Management System

Number of concurrent users	Traditional system response time		B/S architecture system response time	
	Student information addition	Student information modification	Student information addition	Student information modification
20	1.24	1.22	0.81	0.82
40	1.25	1.27	0.86	0.87
60	1.27	1.27	1.01	1.02
80	1.31	1.31	1.12	0.11
100	1.32	1.30	1.03	0.12

different network conditions. Table 3 presents the statistical results of the relevant data.

As shown in Table 3, the system's response time is prolonged with the increase of concurrent users in the system. From horizontal data comparison, it can be concluded that the university student management system designed based on computer BS architecture outperforms the control system in response time for both adding and modifying student information. This indicates that the designed student information classification management system performs better in terms of operational performance.

5 Conclusion

When designing the computer BS mode student information management system, we should analyze the design purpose according to the actual needs, and then realize the design of the functional modules, databases and programs of the system, so as to achieve an efficient system for student information management. The student information management system built on the computer B/S model features a user-friendly interface, supports a wide range of functions, and ensures system security. Its application in college student management enables streamlined and effective handling of student data.

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References

[1] Camargo, R. Y. D., Marozzo, F., & Martins, W. 2021. Computer architecture and high performance computing. Concurrency and Computation: Practice and Experience, 33(18)23.
[2] Yin, X. H. 2021. Construction of student information management system based on data mining and clustering algorithm. Complexity, 2021(2), 1-11.
[3] Yang, H., & Zhang, W. 2022. Data mining in college student education management information system. International Journal of Embedded Systems23(3), 15.
[4] Jasmis, J., Aziz, A. A., Jono, M. N. H. H., Zamzuri, Z. F., & Elias, S. J. 2021. An analysis model for an integrated student activities management system for higher education during rmo/cmco/pasca covid-19 period in malaysia. Procedia Computer Science, 179(2), 798-803.
[5] Feng, S., Zhan, L., & Zhao, H. 2021. Construction and realization of university student information management system based on big data technology. Journal of Physics: Conference Series, 1992(4), 042053-.
[6] Basir, M., Ali, S., & Gulliver, S. R. 2021. Validating learner-based e-learning barriers: developing an instrument to aid e-learning implementation management and leadership. The international journal of educational management23(6), 35.
[7] Sabarudin, A. B., & Razak, F. Z. A. 2021. Understanding the effects of system quality on continuance intention to use student's information system management: an empirical evidence from malaysia. Journal of Physics: Conference Series, 1793(1), 012024-.
[8] Wang, H., Wang, N., Li, M. J., Mi, S., & Shi, Y. Y. 2021. Student physical health information management model under big data environment. Scientific Programming, 2021(1), 1-10.
[9] He, Q., Lu, L., & Xu, X. 2021. University sports information management system based on big data. Journal of Physics: Conference Series, 1881(3), 032058-.
[10] Agarwal, P., Joshi, A., & Naib, B. B. 2021. Research paper on student information system. International Journal of Engineering and Advanced Technology, 10(4), 158-161.